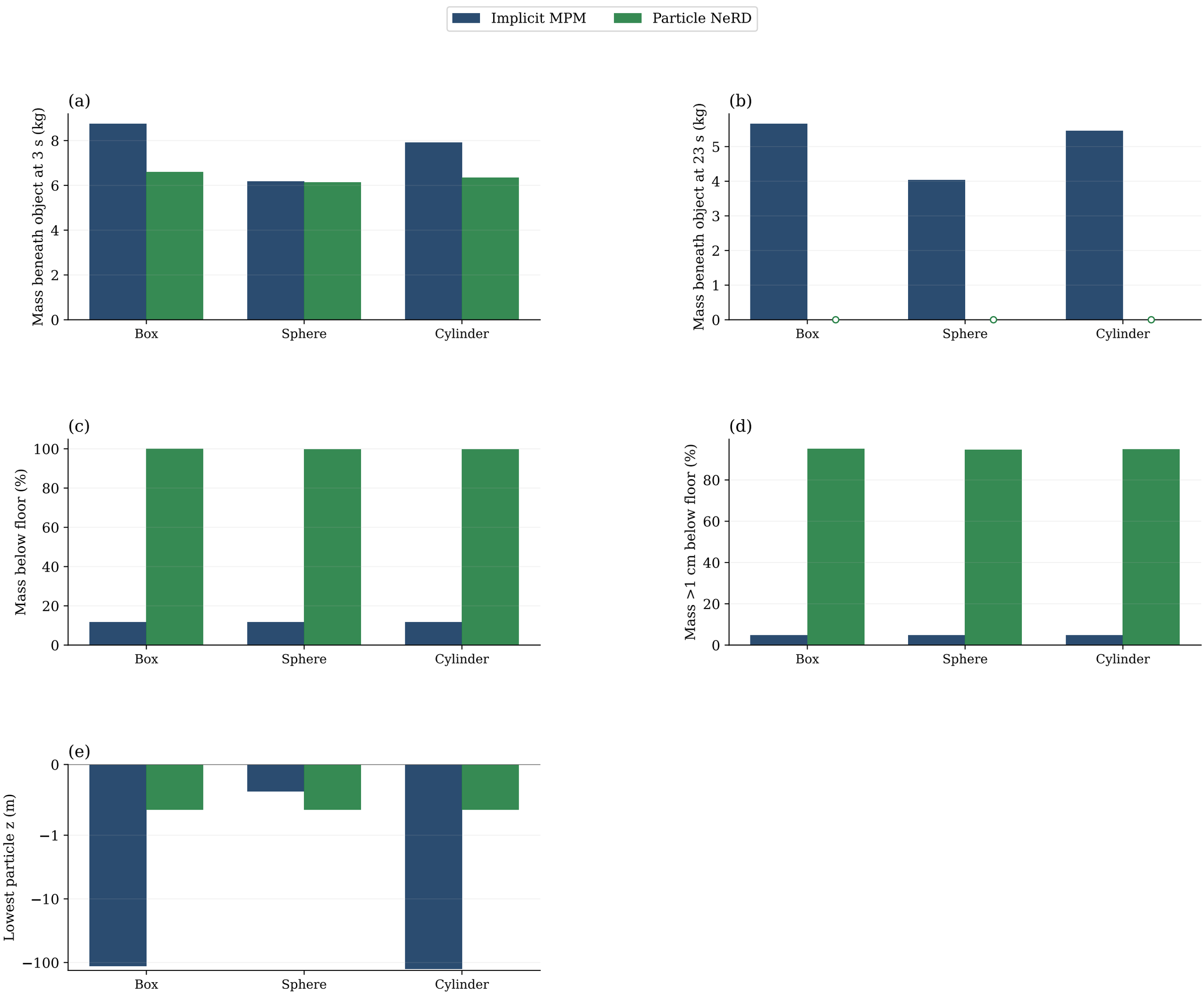

Figure 9: Frozen object-model checkpoint 9,000. Open circles mark measured zeros. (a,b) Physical mass beneath the object at 3 and 23 s; the region is its world x/y bounding rectangle extending 0.10 m below the lowest body point, clipped above the floor. (c) Mass with centers below the z=−0.30 m floor, with 1 µm tolerance; (d) mass more than 1 cm below it; (e) lowest particle-center z at 23 s, with symmetric-log scale (linear within ±1 m). Percentages use total represented physical mass. One paired run per shape; all escaped particles are retained. MPM also settles and leaks.